

Direct Boolean Index Schemata for Scheduling and Scratch Allocation

A reproducible Luminal compiler case study

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Abstract

We study an executable compiler whose scheduling and scratch-allocation decisions are witnesses of exact Boolean-set queries. Integer anchors and free-coordinate masks represent families of candidate decisions; intersections, differences and restrictions operate directly on these families. An independent greedy construction supplies an incumbent, and bounded joint queries combine legality with a strict cycle-scratch product target. We establish the elementary algebra and construction invariants and evaluate a standalone implementation on the pinned Luminal machine. On eight public programs, the combined score is 2.008466 relative to serial, versus 1.901379 for the frozen classical compiler. The public gain already occurs in construction; joint optimization improves six of 100 additional generated programs. Implementation refinements reduce full compilation time by 2.1320 times relative to the previous direct version in the final reviewed run, but compilation remains substantially slower than classical. We report both outcomes, between-run variation, and representation limits. Connections to Shannon decomposition, information bounds and Wolfram’s account of memory retrieval are separated by their actual role. The resulting artifact is a basis for a controlled complexity study, not evidence of a general complexity advantage. The structural-encoding feasibility campaign did not meet its declared coverage gate, leaving all subsequent experiments unmeasured.

1 Question and contribution

Can a compiler construct and improve a schedule by querying exact families of Boolean decision indices, without first obtaining a schedule from a conventional compiler? This case study answers that implementation question on a small, precisely specified machine. The key operation is to intersect the set of legal decisions with an objective threshold and retrieve a witness. Representing that intersection can avoid enumerating individual assignments, when the predicates admit useful structure.

The contributions are a direct integer-mask realization, an independently constructed incumbent, bounded joint schedule/address queries, and a retained evaluation with explicit correctness and evidence gates. Cube algebra and constraint-based compilation have substantial prior art. Our contribution is the particular construction and its measured behaviour, rather than a claim that symbolic sets or integrated compilation are new.

Three costs must remain distinct throughout: the execution cycles of the compiled program, its scratch footprint, and the time and memory used to compile it. The direct method improves the aggregate public output score while spending more compilation time. Even the output improvement is not uniform across programs. An aggregate score cannot replace the component measurements.

2 Machine and objective

The experiment pins the Luminal interview repository at revision 573b8a85 [1]; the bibliography gives the full identifier. Inputs are straight-line operations in SSA dependency order, with consecutive operation identifiers, static buffer ranges and supplied input cases. The compiler must emit every operation exactly once. No instruction rewriting, spilling or extra moves are allowed. Source arithmetic wraps modulo 2^{32} ; schedule and address bookkeeping uses nonwrapping integers.

The scratchpad contains 256 words of 32 bits. Scalars occupy one word; vectors occupy eight consecutive words at an address divisible by eight. Per-cycle issue capacities are two for each of the load, scalar and vector engines, and one for each of the store and flow engines. Latency controls result availability, not how long an issue slot is reserved. If i produces an argument of j ,

$$t_j \geq t_i + \ell_i. \quad (1)$$

Overlapping memory accesses to the same buffer preserve source order when at least one is a store: the later access issues at least one cycle after the earlier. Two loads, or accesses to disjoint ranges, need no such ordering.

For a value v , let $b_v = t_{\text{prod}(v)} + \ell_{\text{prod}(v)}$ be its write cycle. Its lifetime is the *inclusive* interval

$$I_v = [b_v, e_v], \quad e_v = \max(\{b_v\} \cup \{t_j : j \text{ consumes } v\}). \quad (2)$$

An unused result still occupies its write cycle. Reuse requires strict temporal separation because writes precede reads within a cycle. Scratch footprint S is the largest allocated end address, including holes. Scored cycles C are the number of emitted bundles; they need not include the final completion cycle of an unused result.

Let (C_p^0, S_p^0) denote serial metrics for program p . For m programs, the official composite score is

$$Q = \sqrt{\left(\prod_{p=1}^m \frac{C_p^0}{C_p}\right)^{1/m} \left(\prod_{p=1}^m \frac{S_p^0}{S_p}\right)^{1/m}} = \left(\prod_{p=1}^m \frac{C_p^0 S_p^0}{C_p S_p}\right)^{1/(2m)}. \quad (3)$$

For one fixed program, maximizing its score is equivalent to minimizing $P = CS$. One component may worsen while the product improves. This is a declared objective tradeoff, not a claim that memory reductions are execution speedups. Only the eight visible programs are evaluated as the public challenge suite; no private-grader result is available.

3 Direct index schemata

3.1 Semantics and elementary algebra

Use n Boolean coordinates with least-significant-bit-first weights 2^i . Let $U = 2^n - 1$. A cube is $c = (n, a, f)$, with $a, f \in [0, U]$ and $a \& f = 0$. Its anchor a fixes the one bits and its mask f marks free coordinates. Its denotation is

$$\llbracket c \rrbracket = \left\{ a + \sum_{i:f_i=1} \epsilon_i 2^i : \epsilon_i \in \{0, 1\} \right\}. \quad (4)$$

The free-coordinate subset sums are the project's *sumandos*. Freedom is local to a schema; a free coordinate can represent a connected variable. An index encodes an entire assignment of query fields. It is not itself a physical scratch address, although one decoded field may be an address.

The broader project’s *index deconvolution* reconstructs an index family from decimal anchors and their sumandos. Here, queries restrict those compact families before decoding a witness. This is unrelated to inversion of a signal convolution or to a physical memory operation. For $n = 0$, the denotation contains the empty assignment, index zero.

Proposition 1 (Intersection, size and minimum). *For equal-width cubes (n, a, f) and (n, b, g) , the intersection is empty exactly when*

$$(a \oplus b) \& (U \oplus (f \mid g)) \neq 0.$$

Otherwise it is $(n, a \mid b, f \& g)$. A cube has $2^{\text{popcount}(f)}$ members and minimum member a . The minimum of a nonempty finite cube union is the minimum of its anchors.

Proof. An assignment lies in both cubes precisely when it satisfies their fixed coordinates. A conflict occurs at a coordinate fixed differently in both, which is exactly the displayed mask test. In the compatible case a coordinate is free in the intersection only if it is free in both; fixed ones are their union. Each free coordinate has two independent choices, proving the size. All free weights are positive and disjoint from the anchor, so their minimum sum is zero. Taking the minimum over a union proves the last assertion. $\square$

Proposition 2 (Exact disjoint difference). *Subtracting cube d from cube c produces an exact disjoint cover with at most n cubes when they intersect. If they do not intersect, the answer is c .*

Proof. For an intersecting pair, consider coordinates free in c but fixed in d , in ascending order. At each coordinate emit the part opposite to d , and continue with the matching part. Emitted parts have different first mismatches and are disjoint. Every member of $c \setminus d$ has one such first mismatch; the final matching part lies inside d and is discarded. There are at most n split coordinates. With none, c is contained in d and the result is empty. $\square$

Restriction to a coordinate value is intersection with that coordinate’s literal cube. Integer intervals are decomposed into aligned binary blocks and embedded in their fields, with all other coordinates free. Covers may overlap; their sizes cannot in general be added to obtain union cardinality.

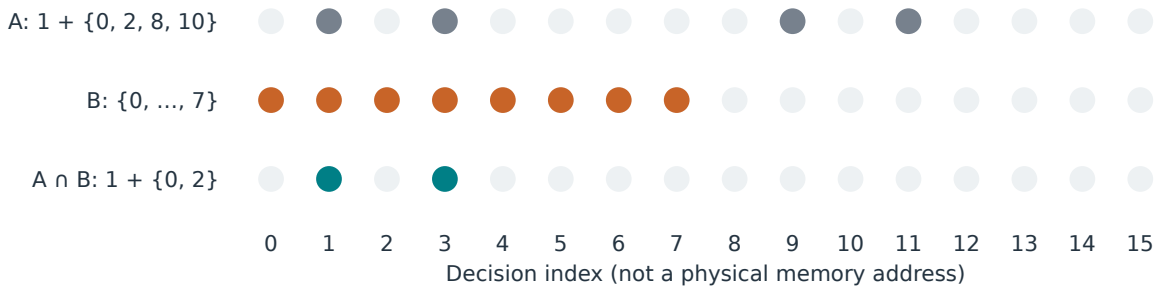


Figure 1: Exact four-coordinate example, with coordinate i weighted by 2^i . Anchor 1 and free mask 10 denote $1 + \{0, 2, 8, 10\}$. Intersecting with indices 0 through 7 leaves $1 + \{0, 2\}$. The dots are a pedagogical expansion, not an enumeration required by the implementation.

3.2 Queries and arithmetic

A query is an AND/OR expression whose leaves are exact cube covers. The solver maintains a current cube and pending predicates. AND adds requirements, OR branches, and a leaf intersects

each alternative with the current cube. Alternatives have deterministic order. A depth-first traversal returns a cube only after every required predicate is discharged. This avoids eagerly forming the Cartesian product of all covers; it does not guarantee a small search.

Proposition 3 (Query soundness and conditional completeness). *If leaves denote their predicates exactly, every returned accepting cube is contained in the expression’s denotation. If the finite search is exhausted without a resource cutoff, absence of an accepting branch proves the expression unsatisfiable in its declared universe.*

Proof. Initially the current cube is the universe. Each leaf intersection preserves exactly the assignments satisfying that leaf. AND requires all children and OR enumerates their union. Induction over the pending expression and search branches preserves these set identities. A completed branch therefore contains only satisfying assignments. Exhausting all branches accounts for the whole expression and proves emptiness if none accepts. $\square$

The anchor of a returned cube is a witness; the first depth-first witness is *not* guaranteed to be the globally smallest index. Bootstrap explicitly uses minimum anchors over complete feasible covers. A timeout or count cutoff is UNKNOWN, never UNSAT. The latter refers only to the current domains and fixed context, not all possible schedules.

Comparison leaves handle fields or constants with integer offsets. On a partial cube, minimum and maximum operand values can settle an inequality for every filling, or exclude every filling. An unresolved comparison splits the free bit of greatest significance within either operand, breaking ties by highest absolute coordinate; zero is explored first. Equality, disequality, constants and identical fields are simplified exactly. All offsets are nonwrapping. This is Boolean cofactoring: the identity

$$F = (\neg x_i \wedge F|_{x_i=0}) \vee (x_i \wedge F|_{x_i=1}) \quad (5)$$

underlies the partition, rather than a Shannon channel-capacity calculation.

4 Compiler construction and bounded improvement

4.1 Independent bootstrap

Let $H_0 = \sum_i \ell_i$. Process operations in source order. For each, form a time interval from its precedence lower bound to a safe upper bound below H_0 , subtract cycles with a full engine, and select the minimum feasible index. The implementation tightens this upper bound using the number of blocked cycles and the prefix latency sum; this preserves the earliest feasible cycle. After scheduling, compute exact lifetimes. Allocate vectors before scalars and sort each group by write cycle and producer ID. Exclude addresses colliding with previously allocated values whose lifetimes overlap, then select the minimum aligned address.

Proposition 4 (Existence of the bootstrap choices). *For a legal nonempty input with positive latencies and total distinct-result width at most 256 words, each scheduling and allocation choice exists, provided query construction completes within its budget.*

Proof. Write $h_i = \sum_{j < i} \ell_j$. Inductively $t_j \leq h_j$. Thus every data predecessor is available by h_i , every ordered memory predecessor issues before h_i , and no earlier operation issues at h_i . Consequently h_i is feasible and $h_i < H_0$. A minimum query cannot choose a later time.

For allocation, suppose k vectors have already been placed. Inductively their end addresses are at most $8k$, since each lowest-address choice could have used the next fresh eight-word block. That block is therefore available for the next vector regardless of temporal overlap. After all vectors, each scalar could use the next fresh word. Reuse can reduce this high-water bound. Total

distinct-result width at most 256, guaranteed for the documented supplied programs, therefore suffices. These are existence arguments, not claims of optimal schedules or footprints. $\square$

The policies are familiar greedy policies: earliest feasible cycle and lowest legal address in a specified order. Their decisions are computed through schema queries. No serial or classical schedule supplies the incumbent, and the direct production path contains neither a BDD backend nor an external solver. Failure during bootstrap emits no partial schedule and invokes no fallback.

4.2 Joint legality and target queries

Select at most four operations. Encode their issue times, produced-value addresses and auxiliary engine lanes in contiguous LSB-first fields, ordered by operation ID. Other decisions are fixed. Selected times lie within two cycles of the incumbent and in $[0, \min(H_0, T) - 1]$; selected addresses range over all aligned starts ending at or below target memory M .

The query intersects domains and targets, data precedence, memory order, engine capacity and scratch safety. Two selected or external operations sharing an engine require different times or lanes. External lanes are fixed by operation ID in each bundle. Since engine capacities are at most two, the pairwise rule is equivalent to issue capacity after existentially quantifying selected lanes. Lanes are not emitted machine instructions.

For every affected value pair u, v , scratch safety requires

$$a_u + w_u \leq a_v \vee a_v + w_v \leq a_u \vee e_u < b_v \vee e_v < b_u. \quad (6)$$

The temporal comparison $e_u < b_v$ expands to $b_u < b_v$ and $t_j < b_v$ for *every* consumer j of u , including consumers outside the window. This exactly implements inclusive lifetimes. Fixed external decisions must also satisfy $t_i < T$ and $a_v + w_v \leq M$; a violation makes the bounded query infeasible. It does not silently expand the window.

For incumbent (C, S) and $P = CS$, try target pairs

$$(C - 1, S), \quad (C, S - 1), \quad \left(C + 1, \left\lfloor \frac{P - 1}{C + 1} \right\rfloor\right), \quad \left(C - 1, \min\left(256, \left\lfloor \frac{P - 1}{C - 1} \right\rfloor\right)\right). \quad (7)$$

Remove undefined, duplicate and out-of-bound pairs, targets below dependency, engine or result-width lower bounds, and any pair with $TM \geq P$. Window priorities consider latest-issued operations and producers of the highest scratch ends, followed by source-order windows of four advancing by two, retaining the final shorter window. For a memory-reducing target, the scratch window comes first. Windows are deduplicated deterministically.

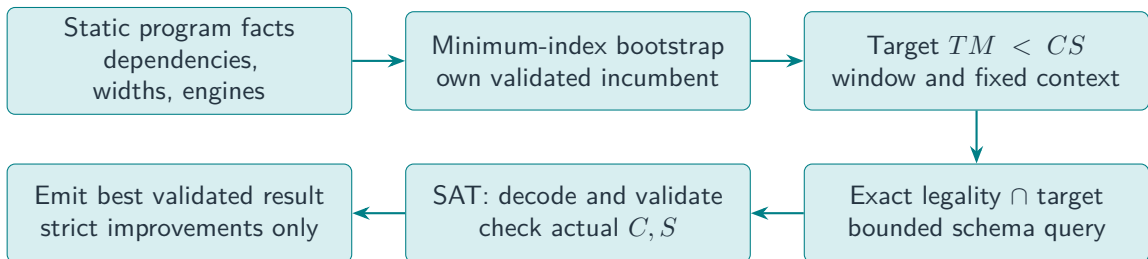


Figure 2: Production flow. UNSAT or UNKNOWN retains the incumbent and continues within the remaining budgets. An accepted improvement restarts target selection. A validator disagreement is recorded as a release-blocking discrepancy. The classical and serial compilers are evaluation controls outside this flow.

Internal validation checks structural machine legality. The evaluation harness separately checks final memory for every supplied case of each measured output. The optimizer measures actual

Listing 1: Abstract optimizer loop; declarations above fix its search policy.

```
incumbent = independently_construct_and_validate(program)
while time_and_query_allowance_remain:
    improvement = false
    for target, window in deterministic_queries(incumbent):
        result = build_and_solve(legality AND target, shared_budget)
        if result is SAT:
            candidate = decode(result.cube.anchor)
            independently_validate_and_measure(candidate)
            record_discrepancy_if_invalid_or_target_breached()
            if valid AND within_target AND product_strictly_decreases:
                incumbent = candidate; improvement = true; restart
            # UNSAT and UNKNOWN preserve incumbent
        if not improvement: break
return incumbent
```

cycles and footprint against the requested target; passing machine validation alone is insufficient. Only strict product decreases are accepted. Conditional on exact constraints and the independent checks, legality is preserved and accepted objective values decrease. Neither property proves global optimality. The test corpus checks the implementation; it is not a formal proof of the full Python program or correctness on untested cases.

4.3 Resource contract and implementation refinements

The declared total soft allowance is 15 seconds, with at most 10 seconds for optimization, 0.1 seconds per joint query and 32 attempted queries. Per-query caps are 4,096 cubes per atomic cover, 50,000 visited cubes and 20,000 expression records. Construction and search share a meter. One second of internal headroom is reserved for final processing; the external isolated-process timeout is 20 seconds. Cooperative clock checks are not hard real-time guarantees.

The refined v4 implementation reduces repeated invariant checking inside proved algebra operations, reuses exact relation covers within one compilation, avoids repeated calendar reconstruction, and streamlines query accounting. Public cube construction still validates inputs. Reuse is charged according to the declared logical accounting; it does not increase the search allowance. Later repairs restore clock checks before terminal verdicts and make the evidence checker enforce its own frozen protocol. These optimizations target constant costs in this implementation, not a different asymptotic problem.

5 Evaluation design and evidence

The primary dataset is the final independent lead run at reviewed revision [1cf98da](#), recorded on 20 September 2026. Production code is unchanged from the preceding repair. The candidate export has 2,652 lines; its full hash and that of frozen v3 are in [Appendix A](#). Earlier runs are retained to describe variability and repairs, not pooled as independent samples.

The fresh verifier passed 340 direct tests, the unchanged 11 public tests, 142 isolated programs with 277 cases, and eight CLI checks. The acceptance corpus combines eight public programs, 30 historical regression programs, 100 generated inputs and four stress inputs. Exact small-domain algebra/constraint tests, independence checks and injected harness failures supplement output tests. The official serial/classical/direct comparison retains all $8 \times 3 \times 3 = 72$ measurements and passes its seven release gates. The evidence audit passes 50 checks covering protocol, membership, hashes, measurement arithmetic and verification artifacts.

For timing, five arms are measured: classical, frozen-v3 bootstrap, frozen-v3 full, candidate bootstrap and candidate full. Each public program has 15 repetitions, giving 600 rows. A separately frozen set of 100 additional generator inputs has three repetitions in each arm, giving 1,500 rows. The latter set uses different seeds but shares the development generator; it is not an independent sample of real compiler workloads.

Each timed compilation uses a fresh process. Direct arms load only their standalone export and pinned machine in an isolated directory. Arm order within each program/repetition is seeded and shuffled. There are no warmups, excluded outliers or silent retries. The compiler-call interval excludes import and the harness’s post-call correctness check. Internal validation performed by the compiler remains inside its call time. Import time, total subprocess elapsed time and process peak resident memory are separate measurements. Process launch paths differ between classical and direct arms, so their process totals also include different harness overheads.

For baseline B and candidate A , the primary speedup statistic is

$$R = \exp \left[\frac{1}{8} \sum_{p=1}^8 \log \frac{\text{median}_r t_{B,p,r}}{\text{median}_r t_{A,p,r}} \right]. \quad (8)$$

The 95% percentile intervals use 10,000 paired resamples of repetition identifiers with seed 20260920. They condition on the programs and recorded run; they quantify neither uncertainty over new programs nor between-run system variation. A ratio of pooled medians is a different statistic and is explicitly labelled wherever used.

The recorded environment is Python 3.13.7, macOS 26.6.2, arm64, with 28 logical CPUs reported. The provenance does not identify the precise CPU model or control thermal state. No verification or benchmark process launched by the lead ran concurrently with these measurements. This does not imply complete system quiescence.

6 Results

6.1 Output quality and the optimizer ablation

The public composite scores are serial 1.000000000, classical 1.901379121, and direct 2.008466202. Both direct versions and both direct modes produce the same public cycles and scratch metrics. The 5.6321% composite advantage over classical therefore belongs to bootstrap on this suite. It does not measure the benefit of joint optimization, nor isolate representation from differing greedy heuristics.

Program	Serial		Classical		Direct, both modes	
	C	S	C	S	C	S
scalar_pipeline	18	15	14	6	14	6
scalar_dual_chain	12	9	9	5	8	3
vector_axpy	14	73	10	32	10	24
vector_bitmix	16	89	10	40	10	40
mixed_broadcast	11	42	9	17	10	24
parallel_memory	21	128	14	56	13	40
scalar_selects	20	16	9	6	9	6
vector_reduction	19	137	12	40	12	40

Table 1: Exact public output metrics from retained measurements. C is emitted cycles and S is scratch words. Direct is worse on both components for `mixed_broadcast`; the aggregate gain is not universal.

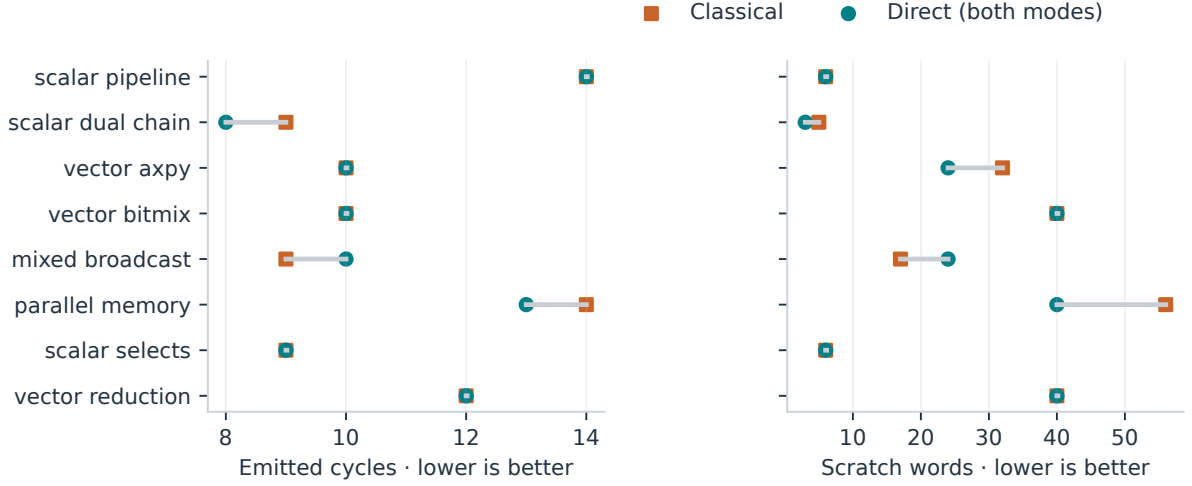


Figure 3: Component differences against classical. Coincident markers indicate equal metrics. The two axes have different units; neither is compiler runtime.

On the 100 additional programs, score geomeans relative to serial are 2.370890261 for classical, 2.362098438 for bootstrap and 2.376596963 for full direct compilation. Six distinct programs improve through joint optimization; 18 accepted improvements are recorded across three repetitions, not 18 independent improved programs. Candidate and frozen direct metrics agree on every public and generated timing row. Bootstrap is slightly below classical on this generated aggregate, while full is slightly above it. The public ordering therefore does not transfer unchanged even to this related generator.

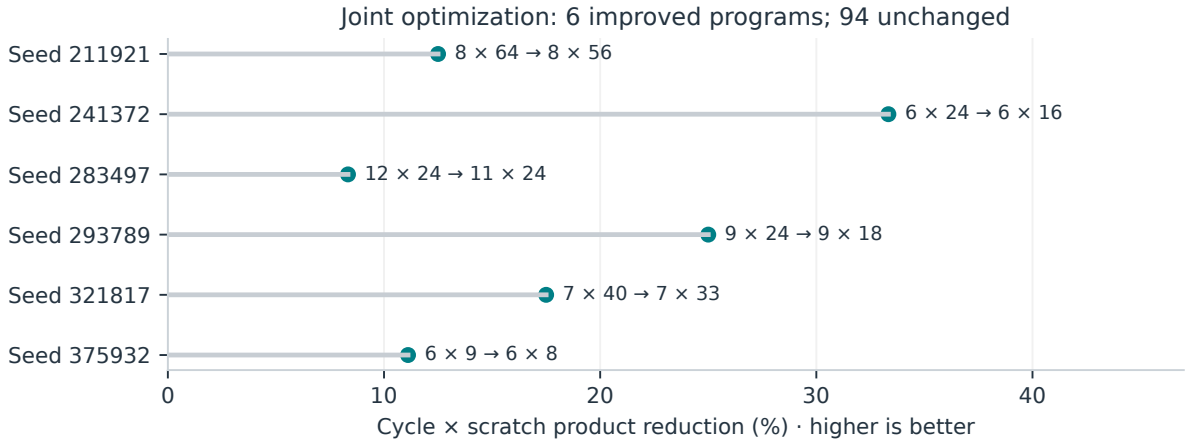


Figure 4: The six additional programs improved by joint optimization, with $(C \times S)$ before and after. The other 94 programs have zero product change. Each pair of metrics repeats in all three measurements, for both direct versions. This is an optimizer-on/off comparison within the direct compiler.

6.2 Compilation time and memory

In the final reviewed run, full-direct compilation is 2.132005693 times faster than frozen v3 by Equation 8, with interval [2.126779601, 2.137538179]. Bootstrap is 1.162943978 times faster, with interval [1.147124692, 1.168961407]. The full 2.00-times engineering target is met in this run; the bootstrap 1.20-times target is missed. These targets were declared before measurement and are separate from mandatory correctness gates.

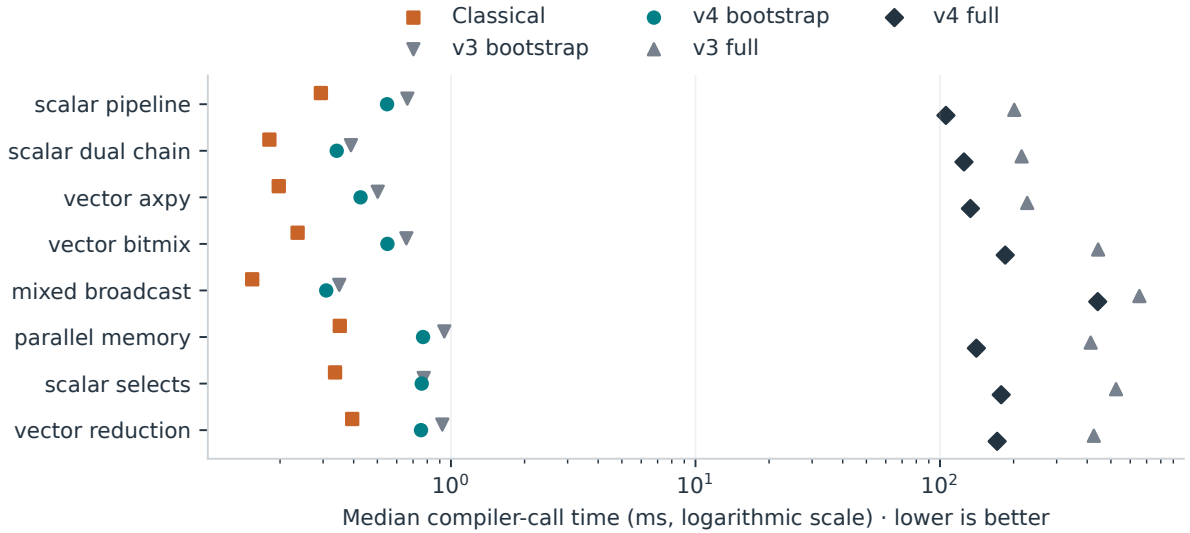


Figure 5: Median compiler-call times from 15 repetitions per public program. The logarithmic axis keeps the classical/direct gap visible alongside the v3/v4 improvements. Bootstrap and full compilation have identical public output quality despite their different costs.

Direct compilation remains slower than classical. The geometric means of per-program median *candidate/classical* ratios are 2.0696 for bootstrap and 657.5045 for full. Alternatively, the *pooled* median call times are 0.26927 ms classical, 0.54962 ms bootstrap and 155.80323 ms full; dividing these pooled medians gives about 2.04 and 578.61. Different aggregation explains the different ratios; none supports classical-relative runtime superiority.

Arm	Call (ms)	Import (ms)	Process (ms)	Peak RSS (MiB)
classical	0.269	0.229	46.305	29.781
frozen bootstrap	0.658	10.886	31.070	22.250
frozen full	417.907	10.841	448.446	24.461
candidate bootstrap	0.550	10.918	31.195	22.281
candidate full	155.803	10.915	185.735	24.914

Table 2: Pooled medians across public timing rows. Peak RSS is whole-process resident memory, including the interpreter and loaded modules; it is not the cube representation size and is unrelated to the program’s scratch words. Each column is separately aggregated, so medians are not additive.

6.3 Variation and negative results

The worker’s last run, using the same production compiler, reported full speedup 1.9471 and bootstrap speedup 1.1337. The lead’s fresh run crosses the full target, but does not erase the earlier miss or establish a reliably achieved 2.00-times speedup. Figure 6 shows consecutive repaired runs. Their within-run intervals are narrow compared with some between-run changes. Two classical program medians changed substantially in earlier rounds; the cause is not established. We therefore retain per-program values rather than diagnosing the difference as hardware noise or attributing it to one edit.

The optimizer improves only a minority of generated inputs, adds no public quality gain, and has substantial overhead. Bounded windows, target selection, representation growth and search all constrain it; the per-query clock is not the only limit. Count-limited queries can become faster even with unchanged budgets. Prior attempts to infer a universal speedup ceiling from timeout fractions are unsupported because queries can encounter different kinds of limits after

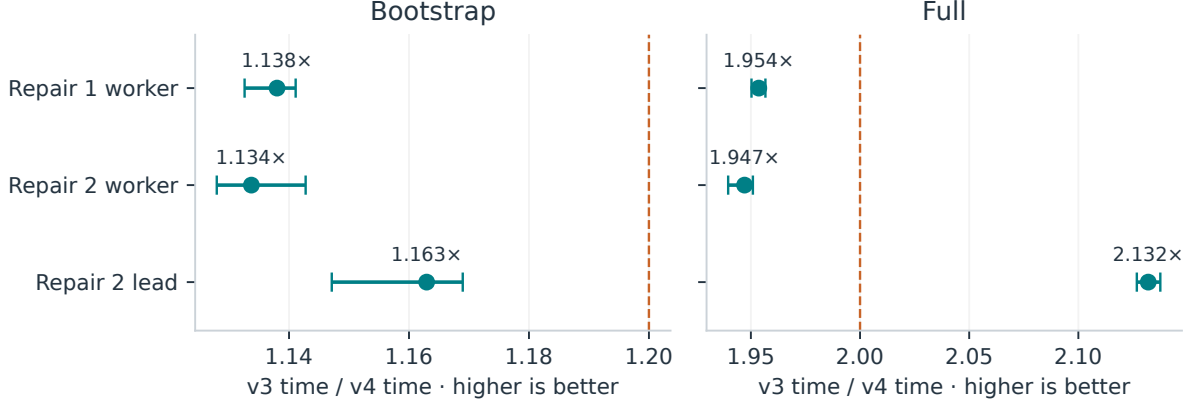


Figure 6: Three retained runs after production repairs. Dashed lines are the original engineering targets. Error bars are paired within-run intervals; they do not describe variation between runs. Production code is the same in these three runs; evidence-checker changes distinguish the repair rounds.

optimization. No measured slowdown here proves that the method can never beat a particular classical implementation.

7 Structural-encoding feasibility

The compiler case study above evaluates one production implementation. A separate, predeclared campaign asked whether exact structural encodings of whole-program schedule/address choices could be sampled and then used in further experiments. This feasibility question matters because rank encodings and structural representations require a well-defined finite domain of legal completions. The campaign’s result is inconclusive: its public sampler did not attain the specified minimum coverage, so the dependent comparisons were not measured.

7.1 Sampling design and finite checks

The locked domain fixes the issue-time fields by operation ID and then the selected scratch-address fields by producer ID, with least-significant-bit-first coordinates. Two distinct streams sampled each public program. Raw-bit sampling draws uniform fixed-width bit strings and decodes them; strings may be invalid or may identify dead ends. Option-path sampling chooses uniformly among the available physical choices at each decision, stopping at a dead end. These are different distributions, and neither is assumed to be uniform over feasible schedules. The domain uses issue cycles from zero through one less than the program horizon and all legal physical scratch bases. The incumbent is first in the option order when available, followed by remaining choices in ascending order. Each stream requested 10,000 attempts per program, with a 60-second per-program, per-stream limit and no invisible retries. Coverage is the union of distinct completed compilations across both streams; the gate requires at least 100 per program. This threshold is a sampling criterion, not an estimate of the total feasible set.

Finite fixture checks covered 12 locked programs and all four declared codecs. Across those fixtures, 24 small code universes were exhausted and 564 round-trip checks were performed. These denominators do not imply that every codec’s complete code universe was exhausted on every fixture. The checks found no round-trip failures or finite-domain discrepancies for the tested objects.

7.2 Public coverage and disposition

Table 3 reports completed draws by stream and the distinct union across streams. Stream counts may include repeated compilations; the union deduplicates them. These are not case-execution counts. All raw-bit streams produced zero completions. The campaign retained 160,000 attempts in 16 streams, 819 completed draws, and 1,356 case checks; it observed zero validator discrepancies. Four programs fell below the 100-completion threshold.

Program	Raw-bit completions	Option-path completions	Distinct union	Gate
scalar pipeline	0	0	0	below minimum
scalar dual chain	0	234	234	meets minimum
vector axpy	0	194	194	meets minimum
vector bitmix	0	55	55	below minimum
mixed broadcast	0	135	135	meets minimum
parallel memory	0	16	16	below minimum
scalar selects	0	168	168	meets minimum
vector reduction	0	17	17	below minimum

Table 3: Phase 2 P1 public sampling coverage. Each stream made 10,000 attempts per program. The union deduplicates compilation identities across streams.

P0 passed its declared provenance, domain and oracle diagnostics. P1 is INCONCLUSIVE because four public programs missed the declared coverage minimum. P2–P5 are blocked by the gate and remain unmeasured. The evidence checker reported zero findings and found the retained artifacts internally consistent, while its record says the evidence is incomplete and scientific success was not established. Checker consistency is not scientific success, and zero observed defects is not a proof of universal correctness. This result does not establish that structural encoding is impossible; it shows that this fixed sampler did not pass its coverage gate. A changed domain or sampling policy requires a separately specified amendment and fresh evidence. The present campaign establishes no compression, scaling, discovery, or performance benefit.

8 Relation to prior ideas and complexity

8.1 Boolean covers and integrated compilation

A cube is a conjunction of fixed literals; a flat cube cover is a disjunctive normal form. Integer anchors change its storage and operations, not its mathematical denotation. Bryant’s ordered binary decision diagrams share subfunctions in a graph and are canonical under a fixed variable order and reduction rules [2]. This implementation instead uses direct masks and factored AND/OR queries. Neither representation dominates all Boolean functions; no BDD performance comparison is reported here.

Combining scheduling and allocation in one constraint model also has precedent. Unison formulates integrated register allocation and instruction scheduling by constraint programming and evaluates the resulting compiler on real target architectures [3]. Luminal’s fixed scratch machine, restricted transformations and four-operation neighborhoods define a much narrower study. Neither its scores nor its times are comparable to Unison’s published results. The present measurements compare only the pinned local implementations.

8.2 What is used from Shannon, and what is not

Equation 5 supplies the mathematical cofactor partition used by Boolean splitting. Shannon’s switching-circuit work is relevant historical context [4]. This identity does not bound the number of splits needed to construct a useful accepting cover.

Two information questions must be distinguished. Shannon’s communication theory concerns source entropy and reliable communication under a specified channel model [5]. A finite memoryless source has entropy $H(X) = -\sum_x p(x) \log_2 p(x)$; a discrete memoryless channel has capacity $\mathcal{C} = \max_{p(x)} I(X; Y)$. The Luminal compiler defines no noisy channel, code rate or source distribution over its decision assignments. It therefore does not implement or approach a measured Shannon communication limit.

Separately, there are 2^{2^n} subsets of an n -bit universe. A fixed-length lossless description capable of distinguishing *all* these subsets needs at least 2^n bits in the worst case, by counting distinct codewords. For variable-length descriptions, fewer than 2^n bits cannot describe all subsets either, since there are only $2^{2^n} - 1$ binary strings shorter than 2^n . This elementary counting argument concerns universal representation capacity, not the running time of one compiler on a structured family. Storing large families compactly is possible precisely because many families are structured.

8.3 Wolfram’s memory discussion

Wolfram discusses exact retrieval by ordered lookup and hashing in his account of human thinking, and suggests a possible connection to memory [6]. The useful motivation here is to ask what an indexed representation lets a query recover without scanning all represented objects. The mechanism in this compiler is exact logical restriction and witness extraction. It is not a model of human memory, cellular-automaton hashing or approximate associative recall. Ordinary implementation caches and provenance hashes do not establish such a connection. The conceptual influence belongs in the research motivation, while claims about computational behaviour must follow from the algorithms and data.

8.4 An explicit representation limit

Proposition 5 (Parity needs an exponential flat cube cover). *For $n \geq 1$, every exact cube-union representation of odd parity requires 2^{n-1} singleton cubes.*

Proof. Any cube with a free coordinate contains two assignments differing only in that coordinate. They have opposite parity. Such a cube cannot be a subset of the odd-parity set. In an exact union, each cube must be a subset, since a union cannot cancel an unwanted member. Every cube is therefore a singleton, and there are 2^{n-1} odd-parity assignments. $\square$

A cube with k free coordinates describes 2^k indices using two n -bit masks, apart from metadata. This is a representation saving, not 2^k answers computed at constant total cost. Expanding those members takes at least their output size. For arbitrary n , Python integer operations themselves scale with the number of machine words; treating an unbounded integer as a constant cost object would hide complexity. Lazy factoring can avoid a large flat cover on some predicates but does not imply tractable search in general. The parity example also shows why the chosen representation must be named in every compression claim.

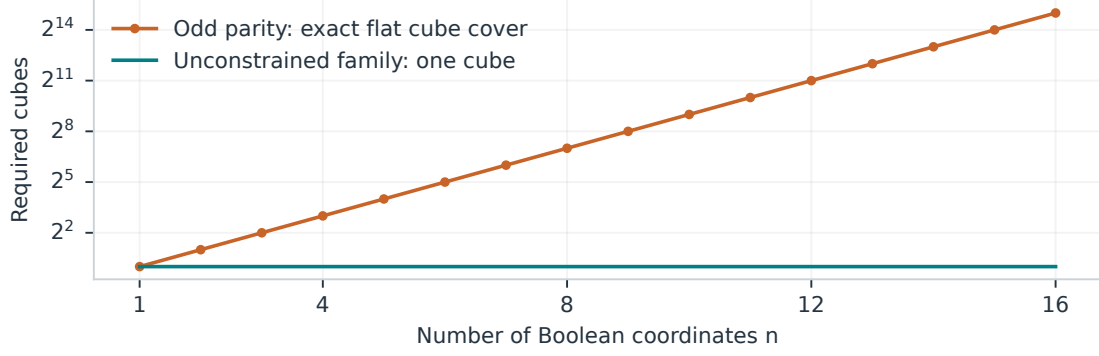


Figure 7: Analytic representation examples, not compiler scaling measurements. An unconstrained family is one cube; an exact flat cover of odd parity needs exponentially many. This lower bound is specific to flat cube unions and does not prove the same growth for factored expressions or decision diagrams.

9 A reproducible starting point for a complexity study

The artifact supports a precise next question: for which families of compiler constraints do direct schemata reduce total construction and query cost? Answering it requires families with controlled growth in operation count, dependency width, field width, alias density and vector pressure. Report expression size, cover sizes, bit widths, search visits, UNKNOWN rates, compile time and process memory separately. Include preprocessing and cache construction in the cost model. Matched-heuristic implementations would isolate representation effects: hold operation order, allocation order, windows, targets and budgets fixed, and vary only the query backend. Stronger compiler baselines and different generators address separate questions of competitiveness and generalization. Repeat measurements across fresh runs and machines rather than treating repeated timings of eight programs as independent workloads. Adversarial predicates, including parity for flat covers, should accompany cases designed to compress.

No such scaling experiment is claimed in this draft. The present evidence supports exact structured operations, validated bounded optimization and a measured quality/cost tradeoff. It does not establish global optimality, polynomial worst-case complexity, universal compression, or a separation of complexity classes. The combination of positive and negative results makes the compiler useful as an instrument for those more narrowly specified experiments.

A Artifact and reproduction

The paper directory contains the editable manuscript, figure generator, generated PDF/PNG figures, tables and an input hash manifest. All numerical figures are derived from retained evidence; the set and parity figures are labelled mathematical illustrations. No benchmark was rerun to select a better paper result. The primary evidence root, relative to `luminal-challenge`, is

```
results/direct_index_v4_optimization_repair2/lead_review/
```

The authoritative files are `final/runs.json`, `final/provenance.json`, `verification/summary.json`, `comparison/runs.json` and `measurement_audit.json`. The frozen-v3 export and evaluation corpus are under `results/direct_index_v4_optimization/baseline/`. The candidate and frozen export SHA256 digests, respectively, are

```
d14bf39b450aaa5fe73c41a3236980e219089e527ae43007781d78e0059a5864
c0574395d339dae3b24e819955795c2a6165953c5966e46062c9bf9b18a50ce2
```

Figure generation checks current production hashes against retained provenance, the candidate export, timing-row membership, component equality between versions, and recomputed scores and speedups. The artifact README gives build commands and validation results. To repeat the full empirical protocol, use new output directories and the commands in the final lead review. Never overwrite the historical measurement trees. The standalone compiler needs Python 3.10+ and the supplied machine module; plotting additionally uses NumPy and Matplotlib, and the paper uses a standard pdfL^AT_EX toolchain.

The separate Phase 2 evidence root is `results/phase2_structural_encoding/phase2_repair_20260923c/`. Its retained acceptance, manifest, gate and hypothesis records, checker output, and P0/P1 summaries are the sources for the feasibility section; `paper/generated/phase2_metrics.json` records their source digests and derived denominators. The P1 raw-attempt file is approximately 113 MB and remains local. Packaging and a portable storage representation are unresolved, so this evidence is not described as portable, published or fully archived. The production measurements above remain the accepted 20 September evidence; the Phase 2 checks are dated 23 September and are not pooled with them.

B Per-program compiler-call medians

Program	Classical	v3 boot	v3 full	v4 boot	v4 full
	Milliseconds, median of 15 calls				
scalar_pipeline	0.2937	0.6632	201.2134	0.5479	105.7414
scalar_dual_chain	0.1809	0.3899	215.6925	0.3410	125.3288
vector_axpy	0.1976	0.5017	227.3233	0.4268	133.1782
vector_bitmix	0.2360	0.6570	443.2952	0.5497	184.8406
mixed_broadcast	0.1538	0.3492	653.8342	0.3089	441.6796
parallel_memory	0.3510	0.9393	413.2939	0.7688	140.8535
scalar_selects	0.3357	0.7737	524.8696	0.7593	177.9620
vector_reduction	0.3945	0.9218	425.7438	0.7541	171.2852

Table 4: Unrounded raw timings are retained in the JSON evidence. These medians underlie Equation 8 and Figure 5.

C Review status and assistance

The implementation received local lead acceptance with limitations on 20 September 2026. That disposition covers the recorded compiler and evidence; it is separate from manuscript peer review or a novelty determination. The draft was prepared with Codex assistance; Claude Code assisted implementation and repair work recorded in the repository. Checks include small-domain oracles, independent machine execution, injected failures, source hashes, recomputation of recorded measurements and inspection of primary literature. The author retains responsibility for reviewing the proofs, claims and interpretation.

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